PRODUCTION DEPLOYED • SUB-30S PRESCRIBING • META WHATSAPP CARE

RxNXT CLOUD

Intelligent Outpatient Operating System & Feature Master Deck

A Unified Clinical Command Center for Doctors, Receptionists, Patients & In-Clinic Pharmacies

DOCTOR

Doctor Workspace

Sub-30s prescribing, additive fuzzy search, 1-click dosage chips & cloning.

WHATSAPP

WhatsApp Care

Meta Cloud API v20.0 with 3-window Smart Slots & Day 25 chronic refill alerts.

QUEUE

Queue & DPDP

0.5s rapid registration auto-reset, sequential tokens, and DPDP 2023 compliance.

PHARMACY

Pharmacy Counter

Unified Rx ID workflow, real-time stock deduction, batch/expiry controls & receipt logs.

Clinical Pain Points & Stakeholder Value Matrix

Unifying the 4 critical OPD touchpoints to eliminate friction, errors, and revenue leakage.

Doctor

FRICTION SOLVED:

Slow paper writing, illegible handwriting risks, non-compliant follow-ups.

KEY DELIVERABLES:

- Sub-30s prescribing engine with 1-click chips
- MCI registration badge & digital signature
- 1-click repeat prescription cloning
- 85% time savings per patient

Receptionist

FRICTION SOLVED:

Waiting room congestion, manual tokens, slow patient data intake.

KEY DELIVERABLES:

- 0.5s auto-clearing registration interface
- Sequential live tokens with waiting timers
- DPDP Act 2023 WhatsApp consent logging
- 70% drop in front-desk chaos

Patient

FRICTION SOLVED:

Lost paper prescriptions, forgotten dose hours, missed chronic refills.

KEY DELIVERABLES:

- Instant WhatsApp PDF delivery + summary
- 3-Window Smart Slot dose nudges (IST)
- Day 25 chronic refill care alert
- 3x higher medication adherence

Pharmacist

FRICTION SOLVED:

Unreadable handwriting, fake/reused scripts, inventory mismatch.

KEY DELIVERABLES:

- Unified Rx ID lookup directly from screen
- Batch, expiry date & rack location tracking
- Automatic real-time stock deduction
- 0 dispensing errors, live stock audit

High-Performance Multi-Tenant Cloud Architecture

Built for zero server cold starts, rock-solid security isolation, and sub-100ms response times.

Frontend & PWA Edge

- Framework: Next.js 14 App Router
- Language: Full TypeScript type-safety
- Design System: Tailwind CSS + Radix UI
- Mobile Experience: Touch swipe gestures, thumb navigation
- PWA: Service worker offline caching & install banner

PostgreSQL & Multi-Tenancy

- Host: Supabase Mumbai Cloud (ap-south-1)
- ORM: Prisma ORM with connection pooling
- Auth: NextAuth.js JWT session strategy
- Isolation: clinicId scoped boundary on all tables
- Security: Passwords hashed with bcryptjs

PDF & Meta Cloud Gateway

- PDF Engine: jsPDF rendered client-side in DOM
- Zero Server Lag: 0 server CPU load for PDF generation
- WhatsApp Gateway: Meta Graph API v20.0 direct
- Zero Cold Starts: No external Twilio/microservice lag
- Document Upload: Direct base64 PDF upload to Meta

PRODUCTION LATENCY & CAPACITY METRICS

- Supabase Mumbai Query Latency: ~35ms
- Drug Fuzzy Search API: <40ms
- Meta WhatsApp Dispatch: Instant
- PDF Render Time: <150ms

- Multi-Clinic Security: Zero cross-tenant data leakage guaranteed via server-side session authentication.

Sub-30-Second Prescription Generation

Combining additive clinical search, 1-click dosage chips, and historical cloning to save clinical time.

Additive Scoring Fuzzy Drug Search

The search API (/api/drugs/search) implements a clinical additive scoring model:

+100 Pts: Exact Clinical Alias Match
- Typing "PCM" short-circuits instantly to Paracetamol 650mg.

+50 Pts: Doctor Prescribing Preference
- Tracks doctor prescribing frequency and ranks favorite medicines first.

+20 Pts: Clinic Standardized Formulary
- Prioritizes clinic-wide featured medicine inventory.

<15 Pts: Low Confidence Safeguard
- Triggers "Did you mean?" UI banner to prevent medication selection errors.

1-Click Dosage Chips & Clinical Cloning

Zero Typing Prescribing Workflow:

- Frequency Chips: [1-0-1] [1-1-1] [1-0-0] [0-0-1] [SOS]
- Duration Chips: [3 Days] [5 Days] [7 Days] [1 Month]
- Meal Instructions: [After meals] [Before food] [At bedtime]

Clinical History Timeline & 1-Click Repeat Cloning:
- Review full encounter history with prior chief complaints & diagnosis.
- 1-Click "Clone Prescription" immediately populates repeat medications.

Regulatory Stamps & Security:
- Dynamic Unified Rx ID (#RX-### / RX-YYMMDD-###).
- Embeds verified NMC doctor registration number & digital signature.

Receptionist OPD Flow & Patient Queue System

Eliminating waiting room friction with sub-second form resets and real-time waiting timers.

0.5s Rapid Intake Form

- Built for high-volume morning OPD queues.
- Form auto-clears in 0.5s after token creation.
- Rapid fields: Name, Phone, Age, Gender, Address.
- Phone duplicate detection pulls past records.
- Zero front-desk lag or manual form resets.

Live Sequential Token Queue

- Issues sequential tokens: Token #14, #15, etc.
- Status states: WAITING, AWAY, SKIPPED, COMPLETED.
- Doctor sees real-time waiting timers ("Waiting 5m").
- 1-Click consultation start directly from queue item.
- Syncs between reception desk and doctor workstation.

DPDP Act (2023) Compliance

- Mandated WhatsApp communication consent checkbox.
- Complies with India Digital Personal Data Protection Act.
- Timestamped consent stored on patient profile.
- Multi-tenant database prevents cross-clinic leaks.
- Patient right to opt out supported.

CLINICAL IMPACT & RECEPTION ROI

- 70% decrease in front-desk telephone enquiries regarding waiting status.
- Eliminates loud calling of patient names, creating a calm, modern clinic atmosphere.

Automated Patient Care & WhatsApp Smart-Slots

Continuous care loop utilizing Meta Cloud API v20.0 with 3-window daily medication nudges.

Ø<ß 8:00 AM IST — Morning S

- Morning doses (1-0-0, 1-0-1, 1-1-1, OD).
- "Take before breakfast / with water".
- Automated doctor follow-up visit reminders.
- Day 25 chronic care refill briefings.

&þ 1:30 PM IST — Afternoon

- Afternoon doses (0-1-0, 1-1-1).
- "Take after lunch with water".
- Zero message spillage: strictly excludes morning and evening-only medicines.

Ø<ß 8:30 PM IST — Night Slot

- Bedtime doses (0-0-1, 1-0-1, 1-1-1, HS).
- "Take after dinner / before bedtime".
- Closes patient daily adherence cycle.
- Prevents missed nighttime antibiotics & sedatives.

Zero Waste & Superseding Rules

When a doctor modifies or re-issues a prescription, all older pending reminders are marked SUPERSE

Day 25 Chronic Refill Engine

For long-term prescriptions (>14 days), fires 5 days before medication ends: "5 days of medicines remaining. Co

In-Clinic Pharmacy Counter & Inventory Engine

Connecting the doctor prescription directly to the dispensary with real-time stock deduction.

Unified Rx ID Dispensing Terminal

- Route: /pharmacist/dashboard
- Synchronized format: #RX-### / RX-YYMMDD-###.
- Pharmacist enters Rx ID or picks patient from dispensing queue.
- Auto-loads prescribed medicines, forms, strengths, and units.
- Prevents handwriting misinterpretation and wrong dispensing.

Real-Time Stock Deduction & Batches

- Route: /pharmacist/inventory
- 1-Click Dispense decrements current stock quantity immediately.
- Tracks Batch Number, Expiry Date, Rack/Shelf Location.
- Highlights low-stock medicines breaching minimum reorder levels.
- Prevents out-of-stock crises and expired medicine dispensing.

DISPENSING LOGS, PAYMENT RECORDING & CLINIC REVENUE RETENTION

- Records payment mode (CASH, UPI, CARD, CREDIT) and total bill amount for each dispense log.

- Tracks unit selling price (MRP) vs cost price for margin calculation and internal clinic audit trail.

Regulatory Verification, RBAC & Multi-Tenant Control

Built-in National Medical Commission (NMC) verification and multi-role clinic operations.

Official NMC REST API

- Live verification against National Medical Commission.
- Checks all 24+ State Medical Councils in ~0.4s.
- Mandates verified license before workspace creation.
- Double validated on server-side NextAuth endpoint.
- Eradicates quackery and unauthorized access.

Role-Based Access (RBAC)

- doctor: Clinical workspace, patients, templates.
- receptionist: Queue tokens, registration.
- pharmacist: Dispensing counter, inventory.
- nurse: Vital check & support workflows.
- admin / superadmin: Clinic profile, billing, staff.

150-Prescription Volume Trial

- 150 free prescriptions on signup (no calendar countdown).
- Smooth onboarding: 0-120 Rx full frictionless access.
- Warning banner at 121-149 Rx; hard lock at #150.
- 1-click self-serve upgrade to Solo GP, Specialist, Polyclinic.
- Non-destructive read-only archive protects clinic data.

CLINIC PROFILE & WHITE-LABELING CUSTOMIZATION

• Drag-and-drop clinic logo upload • Digital signature auto-embedded on PDFs • Custom per-clinic formulary.

Strategic Product Roadmap & Feature Expansions

Horizontal depth into Clinical Decision Support (CDSS), Integrated Billing, and Patient Self-Service.

1. AI Clinical Decision Support (CDSS)

- Drug-Drug Interaction (DDI) alert engine (e.g. Warfarin + NSAID warnings).
- Pediatric & geriatric dose calculations based on age & weight.
- Generic cost-saving switcher: suggests equivalent cheaper formulations.

2. Integrated OPD Billing & Dynamic UPI

- Unified Checkout: Consultation fee + lab + pharmacy bill in one invoice.
- Dynamic UPI QR codes printed on prescriptions and sent via WhatsApp.
- Cash register tracking & daily financial reconciliation reports.

3. Zero-Install Patient Web Portal

- Lightweight phone + OTP login for patients without app install.
- View past prescription PDFs, vitals trends, and appointment bookings.
- Interactive adherence tracker where patients tap "Dose Taken".

4. Permanent Cloud Archival & AWS S3

- Long-term storage of encrypted prescription PDFs on AWS S3 / Supabase.
- SHA-256 cryptographic tamper verification for legal validity.
- Time-limited signed URLs for insurance and audit access.

OPERATIONAL BLUEPRINT COMPLETE

Feature Discussion & Next Steps

Recommended Priorities to Accelerate Pilot Adoption and Clinical Impact

Track A: Clinical Depth

AI Decision Support (CDSS)

- Implement Drug-Drug Interaction alerts.
- Add generic brand switcher.
- Dosage sanity validation.

Impact: Elevates clinical quality & brand trust.

Track B: Monetization

OPD Billing & Dynamic UPI

- Consultation + Pharmacy unified invoice.
- Dynamic UPI QR on prescriptions.
- Daily cash register reconciliation.

Impact: Unlocks direct clinic revenue control.

Track C: Retention

Patient Mobile Web Portal

- Mobile phone OTP authentication.
- Prescription PDF archive on phone.
- Interactive dose tracking check-ins.

Impact: Maximizes patient compliance & returns.